serviceOrchestration

Service Description

Service ID: *"serviceOrchestration"*

Abstract
This document describes the `serviceOrchestration` service, version 1.0.0, produced by the DynamicServiceOrchestration core system of Arrowhead (documentation edition 5.3.0).

Contents

1 Documentation edition and implementation scope **3**

2 Overview **3**
2.1 Significant Prior Art . . . 3
2.2 How This Service Is Meant to Be Used . . . 3
2.3 Important Delimitations and Dependencies . . . 3

3 Service Interface **4**
3.1 operation pull . . . 4
3.2 operation subscribe . . . 4
3.3 operation unsubscribe . . . 4

4 Information Model **5**
4.1 struct AuthorizationTokenGenerationResponse . . . 5
4.2 struct MetadataRequirement . . . 5
4.3 struct OrchestrationNotifyInterface . . . 5
4.4 struct OrchestrationRequest . . . 5
4.5 struct OrchestrationResponse . . . 5
4.6 struct OrchestrationResult . . . 6
4.7 struct OrchestrationServiceRequirement . . . 6
4.8 struct OrchestrationSubscriptionRequest . . . 7
4.9 struct QoSRequirement . . . 7
4.10 struct ServiceInstanceInterfaceResponse . . . 7
4.11 Primitives . . . 8

5 References **9**

6 Revision History **10**
6.1 Amendments . . . 10
6.2 Quality Assurance . . . 10

1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `serviceOrchestration` service, which provides service orchestration: a consumer asks which provider it should use for a service, and receives the matching, authorized providers with the interfaces to call. It is produced by the DynamicServiceOrchestration system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

Successor of the Arrowhead 4.x Orchestrator `orchestration` service (store-based and dynamic orchestration were options of one system; in 5.x they are two systems with the same service definition).

2.2 How This Service Is Meant to Be Used

A consumer calls `pull` with a service requirement and gets the results immediately. Alternatively it subscribes with `subscribe` and an endpoint; the orchestrator then pushes results when triggered, and `unsubscribe` ends the subscription.

2.3 Important Delimitations and Dependencies

Results depend on what the ServiceRegistry, ConsumerAuthorization and (optional) Blacklist, DeviceQoSEvaluator, TranslationManager and Gatekeeper report. The service only selects providers; it does not proxy the consumption. Inter-cloud orchestration through the Gatekeeper is a local extension of this workspace (`enable.intercloud`).

3 Service Interface

This section describes the interfaces to the `serviceOrchestration` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.



Figure 1: SysML-style block of the `serviceOrchestration` service and its operations.

The following interface operations are available.

3.1 operation pull (OrchestrationRequest) : OrchestrationResponse

Executes a dynamic orchestration for the requester: service discovery in the ServiceRegistry, removal of locked, blacklisted and unauthorized candidates, interface matching (optionally with translation), preferences, QoS evaluation, matchmaking, exclusive reservation and token generation. Without local results and with `ALLOW_INTERCLOUD`, or directly with `ONLY_INTERCLOUD`, the Gatekeeper is asked (local extension).

3.2 operation subscribe (OrchestrationSubscriptionRequest) : String

Creates (or replaces) a push subscription of the requester for itself; with `trigger=true` the orchestration is executed and pushed immediately. Returns the subscription identifier, 201 for a new subscription.

Optional parameter: `trigger`.

3.3 operation unsubscribe (Identifier)

Removes a subscription of the requester. Returns 200 if removed, 204 if it did not exist.

4 Information Model

Here, all data objects that can be part of the `serviceOrchestration` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.11, which are used to represent things like hashes and identifiers.

4.1 struct AuthorizationTokenGenerationResponse

A generated token.

Field	Type	Description
tokenType	AuthorizationTokenType	Token type: `TIME_LIMITED_TOKEN`, `USAGE_LIMITED_TOKEN`, `SELF_CONTAINED_TOKEN` or `TRANSLATION_BRIDGE_TOKEN`.
targetType	AuthorizationTargetType	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
token	Token	The token (opaque string, Base64 or JSON Web Token).
usageLimit	Number	Number of allowed uses (usage-limited tokens).
expiresAt	DateTime	UTC expiry of time-limited tokens.

4.2 struct MetadataRequirement

A JSON object of metadata requirements: each key is a metadata path (dot notation), each value the required value or an operator object such as `{"op": "LESS_THAN", "value": 10}`.

4.3 struct OrchestrationNotifyInterface

Notification endpoint of push orchestration.

Field	Type	Description
protocol	String	Protocol of the notification: `HTTP`, `HTTPS`, `MQTT` or `MQTTS`.
properties	Map<String, String>	For HTTP(S) `address`, `port`, `method`, `path`; for MQTT(S) `topic` (and broker `address`, `port`).

4.4 struct OrchestrationRequest

An orchestration request.

Field	Type	Description
serviceRequirement	OrchestrationServiceRequirement	Requirements on the service the consumer wants to use.
orchestrationFlags	Map<String, Boolean>	Orchestration flags, e.g. `MATCHMAKING`, `ONLY_PREFERRED`, `ALLOW_INTERCLOUD`.
qualityRequirements	List<QoSRequirement>	Quality-of-service requirements (evaluated by DeviceQoSEvaluator).
exclusivityDuration	Number	Requested exclusive reservation of the provider in seconds.

4.5 struct OrchestrationResponse

The orchestration result.

Field	Type	Description
results	List<OrchestrationResult>	The orchestration results, best first.
warnings	List<String>	Warnings of the orchestration, e.g. `autoMatchmaking`, `notExclusive`, `interCloud`.

4.6 struct OrchestrationResult

One provider of the requested service.

Field	Type	Description
serviceInstanceId	Identifier	Service instance identifier `<System>|<serviceDefinition>|<version>`.
cloudIdentitifer	Name	Cloud identifier of the provider (`LOCAL` for the own cloud). The JSON name keeps the upstream spelling.
providerName	Name	Name of the provider system.
serviceDefinitition	Name	Service definition name. The JSON name keeps the upstream spelling.
version	Version	Version of the service instance.
aliveUntil	String	Time until which the service instance is registered (its expiry), if any.
exclusiveUntil	String	Time until which the instance is reserved exclusively for the consumer, if reserved.
metadata	Metadata	Metadata of the service instance.
interfaces	List<ServiceInstanceInterface Response>	Interfaces the consumer can use (only matching ones).
authorizationTokens	Map<String, Map<String, AuthorizationTokenGeneration Response>>	Authorization tokens generated for the consumer, per interface template and operation.

4.7 struct OrchestrationServiceRequirement

Requirements on the requested service.

Field	Type	Description
serviceDefinition	Name	Service definition name.
operations	List<String>	Service operations the consumer wants to use.
versions	List<Version>	Accepted service versions.
alivesAt	DateTime	Only instances that are still registered (not expired) at this UTC time are matched.
metadataRequirements	List<MetadataRequirement>	Requirements on the service instance metadata; each map is one alternative.
interfaceTemplateNames	List<String>	Accepted interface templates, e.g. `generic_https`.
interfaceAddressTypes	List<String>	Address types the provider interface must offer.
interfaceProperty Requirements	List<MetadataRequirement>	Requirements on the interface properties; each map is one alternative.
securityPolicies	List<String>	Accepted interface security policies.
preferredProviders	List<String>	Provider systems to prefer.

4.8 struct OrchestrationSubscriptionRequest

A push orchestration subscription.

Field	Type	Description
targetSystemName	Name	Consumer system; default the requester.
orchestrationRequest	OrchestrationRequest	The orchestration request executed for the subscription.
notifyInterface	OrchestrationNotifyInterface	Where and how the orchestration result is pushed to the target system.
duration	Number	Requested lifetime of the subscription in seconds; no value means unlimited.

4.9 struct QoSRequirement

One quality-of-service requirement.

Field	Type	Description
type	String	Type of the QoS evaluation, as known by DeviceQoSEvaluator.
operation	String	QoS operation: `FILTER` or `SORT`.
requirements	Map<String, Object>	Parameters of the QoS requirement.

4.10 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

Field	Type	Description
templateName	Name	Name of the interface template, snake_case (e.g. `generic_https`).
protocol	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
policy	String	Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies.
properties	Map<String, Object>	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

4.11 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
AuthorizationTargetType	One of `SERVICE_DEF`, `EVENT_TYPE`.
AuthorizationTokenType	One of `TIME_LIMITED_TOKEN`, `USAGE_LIMITED_TOKEN`, `SELF_CONTAINED_TOKEN`, `TRANSLATION_BRIDGE_TOKEN`.
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.
Token	An opaque string used as credential.
Version	Specifies a version, semantic versioning `major.minor.patch`.

5 References

[1] J. Delsing, “DynamicServiceOrchestration – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, serviceorchestration-dynamic/documentation/v5.3.0/DynamicServiceOrchestration-v5.3.0-SysD.pdf.

[2] ——, “serviceOrchestration HTTP/TLS/JSON – Interface Design Description (DynamicServiceOrchestration),” Eclipse Arrowhead, IDD 5.3.0, 2026, dynamicServiceOrchestration-v5.3.0-serviceOrchestration-HTTP-IDD.pdf.

[3] ——, “serviceOrchestration MQTT/TLS/JSON – Interface Design Description (DynamicServiceOrchestration),” Eclipse Arrowhead, IDD 5.3.0, 2026, dynamicServiceOrchestration-v5.3.0-serviceOrchestration-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	